A Report on Food Product Development

Company Name: REVITA LTD.

Product Name: ECONUGGS (Bite Size Happiness)

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Abstract :

This report gives a complete description of a new innovative product named EcoNuggs which is made from Banana Peel. Banana peel is considered to be waste though it has high nutritive value. The main ingredient of the nugget is banana peel, additionally, healthy vegetables like cabbage, mushroom, and green peas are added to provide enough nutrients and desirable texture. The Nuggets are seasoned with mixture of spices including chili powder, paprika, black pepper, and salt. Binding agents like Corn Starch and Textured Vegetable Protein (TVP) are used to give a meaty and springy texture and desirable consistency to the nuggets. The entire formulation is completely free from any artificial ingredients, ensuring a healthy eating experience. The natural preservatives ensures a long shelf life of the product that is about the same as nuggets made from protein source. Sensory evaluation yielded positive responses that indicates the product is highly acceptable and desirable for consumers which means the product will have a market demand. Upon proximate analysis the nuggets showed to possess high amount of fiber and protein ensuring the demand for healthy and sustainable food choices.

Chapter 1

Introduction

Nuggets are small, bite-sized food items that are typically breaded and fried, widely recognized for their convenience, versatility, and appeal across various age groups. Traditionally made from chicken, fish, or vegetables, nuggets have become a staple in fast food and home-cooked meals, offering a quick and satisfying meal option. Their adaptability to various flavor profiles and textures makes them an ideal canvas for culinary innovation. The main ingredient in the making of banana peel nuggets is banana peel. Besides that other ingredients include Cabbage, Mushroom, Green peas, Bread crumbs, Textured Vegetable Protein (TVP), Chili powder, Paprika, etc. All of the ingredients adds to the unique flavor and texture of banana peel nuggets making it a unique product.

It is common for the banana peel to be discarded in spite of having a high nutritional profile. For the development of our product we choose the variety Pisang Berangan which is very common in Malaysia. Comprising approximately 35-40% of the total weight of a banana, the peel is an excellent source of dietary fiber, essential minerals such as potassium and magnesium, and bioactive compounds with antioxidant properties (Zainuddin et al., 2018). Traditionally used as animal feed or in composting, banana peels have recently gained attention in the food industry as a sustainable and nutritious ingredient that can be up cycled into human food products (Hartati et al., 2022).

The development of banana peel nuggets represents an innovative approach to food production, addressing both environmental and health concerns. By transforming banana peels into a palatable and nutritious food product, this initiative not only reduces food waste but also provides a healthy alternative to conventional meat-based nuggets. The incorporation of banana peel into nuggets aligns with the growing consumer demand for plant-based, eco-friendly products that support sustainable eating habits (Setiawati et al., 2021).

The unique combination of fiber and natural sweetness from the peels gives a very satisfying mouth feel and also enhances the flavor and texture. The presence of other ingredients like cabbage, mushroom, green peas and TVP further improves the texture and adds to the umami taste of the nuggets. The subtle pleasant spicy taste of the nuggets comes from the unique blend of seasoning that includes black pepper, paprika, soy sauce etc. Overall, the nuggets have a good appeal and taste which makes them a different and unique culinary product.

This report explores the potential of banana peel nuggets as a sustainable and nutritious food product. We will examine the nutritional composition of banana peels, the process of converting them into nuggets, and the sensory evaluation of the final product. The study aims to demonstrate how banana peels can be successfully integrated into the human diet, contributing to a more sustainable and health-conscious food system. The use of banana peel in nugget production is not only a testament to the innovative possibilities within the food industry but also a step forward in the global effort to minimize food waste and promote sustainability.

Literature Review:

Banana peel, despite being a waste product has a huge potential in food industry for its unique sets of properties which includes being rich in fiber and presence of high amount of nutrients. . Research indicates that banana peels are abundant in dietary fiber, essential minerals such as potassium and magnesium, and bioactive compounds with antioxidant properties, making them an excellent addition to health-focused food products (Zainuddin, Rahmat, & Zainal, 2018; Hartati, Setiawati, & Nurhayati, 2022). Additionally, banana peels contain phenolic compounds that contribute to their antioxidant activity, which can be beneficial in reducing oxidative stress and potentially lowering the risk of chronic diseases (Happi Emaga et al., 2011). The incorporation of banana peels into food products, such as nuggets, not only enhances their nutritional profile but also aligns with environmental sustainability goals by reducing food waste (Setiawati, Hartati, & Nurhayati, 2021).

Studies prove that there are major functional benefits of banana peel. For instance, banana peels have been found to possess water-holding and emulsifying properties, which can improve the texture and moisture retention in food items like nuggets (Setiawati et al., 2021; Kadam et al., 2012). Furthermore, incorporating banana peels into food products can introduce unique flavors, making them appealing to consumers who are increasingly seeking innovative plant-based alternatives (Kadam, Prasad, & Patil, 2022). The consumer acceptability seems to be the major problem despite having benefits. Educating consumers about the health benefits and environmental impact of banana peel products is crucial for enhancing their marketability (Hartati et al., 2022). Moreover, further research is necessary to explore the long-term health effects of consuming banana peel-enriched foods and to understand consumer preferences across different demographics (Adekomaya & Adekoya, 2020). Despite all the research done on banana peel, there has been no research done based on the presence of Textured Vegetable Protein (TVP) and the market potential of banana peel nuggets. This report delves deep into the role played by TVP on the texture of nuggets. Additionally, this report will also look into the market potential of banana peel nuggets.

Research Objectives

The objectives of the study are:

1. **To analyze** the nutritional composition of banana peels, focusing on dietary fiber, essential minerals, and bioactive compounds.
2. **To develop** a formulation for banana peel-based nuggets, ensuring optimal texture, flavor, and functional properties suitable for consumer acceptance.
3. **To evaluate** consumer perceptions and acceptance of banana peel nuggets through sensory testing and market surveys.

Chapter 2

Materials Method:

The research was carried out at the FPSM processing lab at Universiti Malaysia Terengganu(UMT), Malaysia.

Product name: ECONUGGS.

Raw Materials:

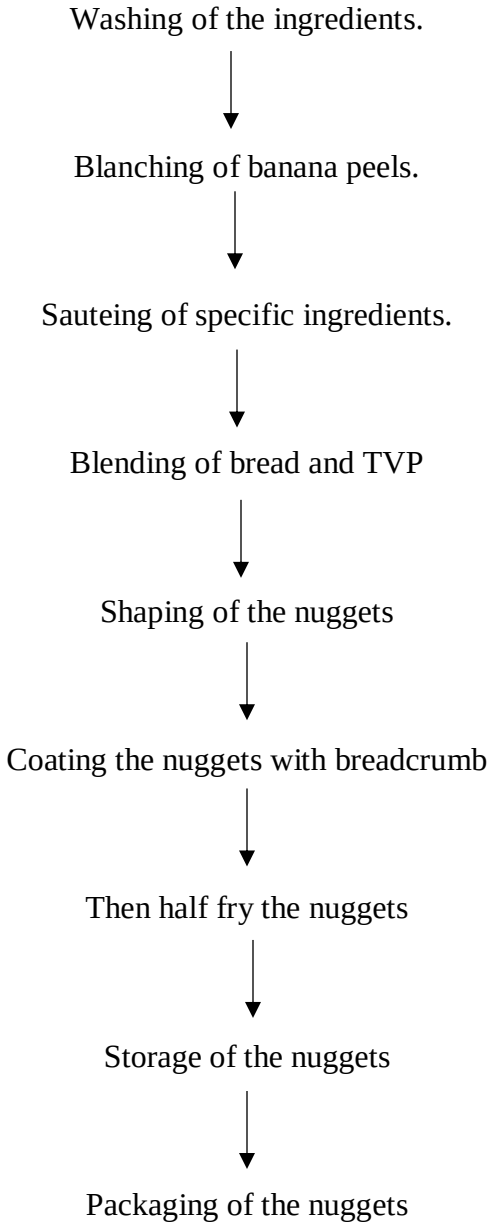
Ingredients	Amount
Banana peel	51.10 gm.
Cabbage	25.55 gm.
Green Peas	25.55 gm.
Mushroom	30.15 gm.
TVP	25.55 gm.
Onion	38.03 gm.
Garlic	7.21 gm.
Ginger	5.51 gm.
Salt	0.24 gm.
Bread	25.55 gm.
Sauce	6.62 ml.
Spices	4.16 gm.
Bread Crumb	40 gm.
Flour	40 gm.
Cooking oil	60 ml

Procedure: ECONUGGS

1) All the raw materials are washed properly.
2) Then all the ingredients was scaled properly.
3) Blanching the banana peels, cabbage and green peas for 2-3 mins.
4) Then dipped in the vinegar.
5) Sauteing onion, garlic, ginger and mushroom
6) Blending all the raw ingredient including bread and TVP
7) Adjusting Salt
8) Forming the shape
9) Dip in to the batter and coat with breadcrumb
10) Half fried the nuggets with oil for 2 mins.

11) Packaging the half fried nuggets.
12) Store at -18° C.

Processing Flow:



Functional properties of the ingredients

A) Banana Peel:

Banana peel is a great source of dietary fiber, adding bulk and improving the texture of the nuggets. Its water-binding ability helps retain moisture, ensuring the nuggets remain tender. Additionally, banana peels provide vitamins and antioxidants, contributing to the nutritional value. The fiber adds slight chewiness and structure to the nugget.

B) Cabbage:

Cabbage contributes moisture to the nugget mixture, keeping it juicy and soft. It also adds volume and a light crunch when chopped finely, enhancing the texture. Cabbage is rich in vitamins, boosting the nugget's nutritional profile, and can help with binding when softened.

C) Green Peas:

Green peas are an excellent plant-based protein source, adding nutritional value to the nuggets. Their mild sweetness balances the overall flavor, and their firm yet creamy texture enhances the bite. Mashed peas also help in binding the ingredients together.

D) Mushroom:

Mushrooms provide a rich umami flavor, enhancing the savory taste of the nuggets. With their high water content, they keep the nuggets moist and juicy. They also contribute a meaty, chewy texture and add important vitamins and minerals.

E) Textured Vegetable Protein (TVP):

TVP boosts the protein content and adds a meat-like texture to the nuggets. When hydrated, it helps bind the ingredients together, giving the nuggets a firm structure. TVP also absorbs fats, helping to retain flavor and moisture during cooking.

F) Breadcrumb:

Breadcrumbs act as a binding agent, absorbing moisture and keeping the nuggets intact. They also create a crispy outer layer when used for coating. Additionally, breadcrumbs add bulk and help balance moisture within the nugget mixture.

G) Bread:

Bread, often crumbled or soaked, helps retain moisture in the nugget mixture. It acts as a binder by absorbing liquid and holding the ingredients together. Bread also softens the texture, making the nuggets tender and less dense.

H) Flour:

Flour acts as a binding agent, helping to hold all the ingredients together. It also contributes to the structure and firmness of the nuggets. Additionally, flour absorbs excess moisture, preventing the nuggets from becoming too wet or falling apart during cooking.

I) Salt:

Salt enhances the overall flavor by bringing out the natural taste of the ingredients. It also helps balance sweetness and bitterness, making the nuggets more palatable. Moreover, salt can assist in preserving the nuggets by inhibiting the growth of bacteria.

Chapter 3

Product Characteristics:

Appearance:

Appearance is the first sensory attribute that consumers notice, making it a key factor in food acceptance. The visual evaluation of Banana Peel Nuggets includes color, shape, and surface texture. A consistent golden-brown color after cooking enhances their appeal, signaling proper frying or baking. Evaluators use standardized scales to rate the color uniformity and surface crispness of the nuggets. The incorporation of banana peel powder and vegetables gives the nuggets a unique visual texture, highlighting their wholesome ingredients.

Taste and Flavor:

Taste refers to the basic sensations such as sweet, sour, salty, bitter, and umami. In Banana Peel Nuggets, the taste combines the mild sweetness of banana peel with the savory elements of vegetables and seasonings. Flavor, which is a combination of taste, aroma, and texture, offers a rich sensory experience. The addition of spices and Textured Vegetable Protein (TVP) enhances the umami flavor, while mushrooms and vegetables like cabbage and green peas contribute to a balanced taste profile. Yeast and spices give the nuggets an appealing aroma, and the crispy exterior provides a satisfying texture.

pH:

The pH level of Banana Peel Nuggets ranges from 5.5 to 6.5, influenced by the banana peel and vegetables used. This slightly acidic environment is beneficial for maintaining a mild flavor and preserving the product's freshness.

Fiber:

Unlike traditional nuggets, Banana Peel Nuggets are rich in dietary fiber. The inclusion of banana peel and vegetables like cabbage and green peas significantly boosts the fiber content, promoting digestive health and making the nuggets more filling.

Ingredient Description:

- **Banana Peel**: Banana peel is high in fiber, which adds bulk and improves the texture of the nuggets. It also contains essential vitamins and minerals like potassium and antioxidants. Its fiber helps retain moisture during cooking, keeping the nuggets tender, while adding a subtle sweetness and mild flavor. (Mohapatra, Mishra, & Sutar, 2010)

- **Cabbage**: Cabbage provides moisture and volume to the nuggets. Rich in vitamins like vitamin C and K, it enhances the nutritional profile. When cooked, it softens, adding a light, crunchy texture and aiding in binding the ingredients together.
- **Green Peas**: Green peas are a source of plant-based protein and fiber, contributing to the nutritional content of the nuggets. Their natural sweetness balances the flavors, while their texture adds a slight creaminess to the bite.
- **Mushroom**: Mushrooms are known for their umami flavor, which enhances the overall taste of the nuggets. They also help in moisture retention, keeping the nuggets juicy inside, while their chewy texture mimics the feel of meat. Mushrooms are also rich in vitamins and minerals, particularly B-vitamins and selenium.
- **Textured Vegetable Protein (TVP)**: TVP is a key ingredient for providing structure and protein in the nuggets. When hydrated, it has a meat-like texture that adds body to the nuggets. It also acts as a binder and absorbs moisture, ensuring the nuggets maintain their shape and do not dry out during cooking.
- **Flour**: Flour helps to bind the ingredients together and gives structure to the nuggets. It absorbs moisture, preventing the nuggets from becoming too wet. Flour also helps in forming a crispy outer layer when cooked, enhancing the texture.
- **Breadcrumbs**: Breadcrumbs act as both a binder and a coating. Inside the nugget mixture, they absorb moisture and help hold everything together. When used on the outside, they create a crispy texture after frying or baking.
- **Yeast**: Yeast is used for its leavening properties, making the nuggets light and slightly puffy. It also adds a distinct, mild flavor that enhances the overall taste of the nuggets. Yeast fermentation contributes to the aroma, making the nuggets more appealing.
- **Salt**: Salt enhances the flavor of the ingredients, balancing sweetness and bringing out the savory elements. It also contributes to the texture by strengthening the protein network during cooking.
- **Vinegar**: Vinegar acts as a preservative, extending the shelf life of the nuggets. It also helps tenderize the mixture and contributes to a slight tang, balancing the flavor profile.
- **Butter/Shortening**: Fat from butter or shortening enhances the tenderness and moisture of the nuggets. It also contributes to flavor and helps create a crispy exterior when the nuggets are cooked.

Chapter 4

Product acceptability

Sensory evaluation

The acceptance test was carried out to determine the acceptance of ECONUGGS. The three formulations of ECONUGGS were mainly based on the application of different percentages of Textured Vegetable Protein (TVP). Acceptance test includes parameters of appearance, color, aroma, texture, taste, and overall acceptability. The test was conducted by 20 untrained panelists using the score sheet in a google form with a scale of 1 to 7 which describes the profile of the nuggets that are tested. The hedonic test uses the number 7 for the highest score (like extremely) and the number 1 for the lowest value (dislike extremely). The criteria for assessment of the acceptance test are presented. The sample was served in a small plastic packet and coded with 3-digit codes before it was permuted. The permuted sample of nuggets was then served in a tray with the water and tissue.

Hedonic scale	Value
Like extremely	7
Like Moderately	6
Like slightly	5
Neutral	4
Dislike slightly	3
Dislike Moderately	2
Dislike extremely	1

The Collected Data is shown just below:

Sample	Appearance	Color	Aroma	Texture	Taste	Overall Acceptibility
A	5	4	6	6	6	6
A	7	7	6	7	7	7
A	5	6	6	7	4	6
A	7	7	7	7	7	7
A	6	6	5	5	5	5
A	7	7	7	7	5	6
A	6	6	6	6	6	5

A	5	5	6	4	5	6
A	2	5	5	6	6	6
A	4	6	6	6	6	6
A	4	5	6	2	2	3
A	4	4	5	4	4	4
A	6	5	6	5	5	5
A	7	7	7	7	7	7
A	4	5	7	4	6	6
A	5	6	5	4	3	4
A	4	4	4	2	4	4
A	5	6	6	4	6	5
A	3	3	4	5	5	4
A	5	5	5	5	5	5
B	5	5	5	6	6	6
B	6	6	5	6	5	5
B	4	3	4	5	3	4
B	5	5	6	4	5	5
B	7	7	7	7	7	7
B	7	7	6	5	6	6
B	6	6	7	7	7	7
B	5	6	4	5	5	6
B	6	5	7	7	7	7
B	5	5	4	5	5	5
B	5	4	4	5	4	4
B	4	4	3	3	2	3
B	4	4	5	5	4	4
B	4	4	5	5	4	4
B	7	7	7	7	7	7
B	6	5	7	5	4	5
B	6	5	6	5	6	6
B	6	6	6	6	6	6
B	6	6	7	6	7	6
B	2	2	4	6	4	4
C	1	1	3	3	3	3
C	3	3	3	1	3	3
C	7	7	7	7	7	7
C	5	6	7	6	7	6
C	6	6	6	7	6	6
C	5	4	6	6	5	6
C	3	3	3	2	3	4
C	2	2	3	3	4	3

C	3	3	3	3	3	3
C	5	5	5	5	5	5
C	4	4	3	5	5	5
C	4	5	5	5	5	5
C	6	6	7	6	6	6
C	6	6	7	6	6	5
C	5	5	5	5	5	5
C	5	4	5	3	5	5
C	6	6	6	6	7	6
C	4	5	4	4	4	4
C	3	3	4	6	6	6
C	4	5	4	5	4	4

Appearance

To measure the appearance acceptability we conducted ONE WAY ANOVA using Minitab.

Considering

A: Banana Peel Nuggets with TVP 50% wt. of Banana Peel

B: Banana Peel Nuggets with TVP 60% wt. of Banana Peel

C: Banana Peel Nuggets with TVP 70% wt. of Banana Peel

Our analysis showed that there is no significant difference of appearance acceptability between usage of different percentage of TVP. (p-value= 0.093)

A≠B≠C

Nuggets Sample	**Appearance**
A	5.05 ± 1.40^{ab}
B	5.30 ± 1.26^{a}
C	4.35 ± 1.53^{b}

Color

To measure the color acceptability we conducted ONE WAY ANOVA using Minitab.

Considering

A: Banana Peel Nuggets with TVP 50% wt. of Banana Peel

B: Banana Peel Nuggets with TVP 60% wt. of Banana Peel

C: Banana Peel Nuggets with TVP 70% wt. of Banana Peel

Our analysis showed that there is no significant difference of color acceptability between usage of different percentage of TVP .(p-value= 0.071)

A≠B≠C

Nuggets Sample	**Color**
A	5.45 ± 1.15^{a}
B	5.10 ± 1.33^{ab}
C	4.45 ± 1.57^{b}

Aroma

To measure the aroma acceptability we conducted ONE WAY ANOVA using Minitab.

Considering

A: Banana Peel Nuggets with TVP 50% wt. of Banana Peel

B: Banana Peel Nuggets with TVP 60% wt. of Banana Peel

C: Banana Peel Nuggets with TVP 70% wt. of Banana Peel

Our analysis showed that there is no significant difference of aroma acceptability between usage of different percentages of TVP. (p-value= 0.065)

A≠B≠C

Nuggets Sample	Aroma
A	5.75 ± 0.91^{a}
B	5.45 ± 1.32^{ab}
C	4.80 ± 1.54^{b}

Texture

To measure the texture acceptability we conducted ONE WAY ANOVA using Minitab.

Considering

A: Banana Peel Nuggets with TVP 50% wt. of Banana Peel

B: Banana Peel Nuggets with TVP 60% wt. of Banana Peel

C: Banana Peel Nuggets with TVP 70% wt. of Banana Peel

Our analysis showed that there is no significant difference of texture acceptability between usages of different percentages of TVP. (p-value= 0.230)

A=B=C

Nuggets Sample	Texture
A	5.15 ± 1.57^{a}
B	5.50 ± 1.05^{a}
C	4.70 ± 1.69^{a}

Taste

To measure the taste acceptability we conducted ONE WAY ANOVA using Minitab.

Considering

A: Banana Peel Nuggets with TVP 50% wt. of Banana Peel

B: Banana Peel Nuggets with TVP 60% wt. of Banana Peel

C: Banana Peel Nuggets with TVP 70% wt. of Banana Peel

Our analysis showed that there is no significant difference of taste acceptability between the usages of different percentages of TVP. (p-value= 0.805)

A=B=C

Nuggets Sample	**Taste**
A	5.20 ± 1.32^{a}
B	5.20 ± 1.47^{a}
C	4.95 ± 1.36^{a}

Overall Acceptability

To measure the overall acceptability we conducted ONE WAY ANOVA using Minitab.

Considering

A: Banana Peel Nuggets with TVP 50% wt. of Banana Peel

B: Banana Peel Nuggets with TVP 60% wt. of Banana Peel

C: Banana Peel Nuggets with TVP 70% wt. of Banana Peel

Our analysis showed that there is no significant difference of overall acceptability between the usages of different percentages of TVP. (p-value= 0.320)

Nuggets Sample	Overall Acceptibility
A	5.35±1.13[a]
B	5.34±1.23[a]
C	4.85±1.23[a]

Interpretation

There is no significant difference in any of the samples for all the parameters mentioned. Mean SD was higher for sample A for all the sensory parameters. So, sample A is mostly accepted.

Chapter 5

Shelf Life

Shelf life is one of the most important aspects of a product. It is an indicator of product acceptability. Any product that passes the time entitled as shelf life will be regarded as unfit for consumption. For food products like Nuggets, it is important to make sure that the shelf life persists for a long. Because being a nutrient-rich food it falls in the region of highly perishable foods. Hence to increase the shelf life some steps are taken. This includes applying various preservation methods including heating the product to adding additives.

For our ECONUGGS the additives that we used includes vinegar and Sodium Benzoate.

Vinegar is a natural preservative that acts by lowering the ph of food which creates an unfavorable condition for microorganism growth. Sodium Benzoate is an effective artificial preservative that is known worldwide for its preservation capability. Per nugget requires 0.1% sodium benzoate.

After using the preservatives it is seen that for our ready to cook ECONUGGS the estimated shelf life will be 3-6 months.

Chapter 6

Proximate analysis:

Proximate analysis is a technique to measure the chemical properties of a product/compound based on five particular elements: moisture content, carbohydrate, protein, fat, and ash. The whole process for each of them is given below:

1. SAMPLE PREPARATION

Sample preparation for Proximate Analysis should be carefully performed to avoid errors in the final result this procedure must involve one or more treatments to homogenize the sample. i.e. the particle size of the sample should ideally be reduced to a size ≤1 mm. Homogeneity of the analytical sample improves the reproducibility of the method and also offers the possibility to reduce the sample size used without sacrificing the quality of the final results. The sample can be homogenized by using a blender, grinder, or sample mill.

1. Determination of moisture content

Method: Oven Drying

Equipment: Oven, 100-110°C, Analytical Balance, Crucible, Desiccator

Procedure:

1. Label and dry crucible in oven at 100-110°C for at least 30 minutes.

2. Cool the crucible in the desiccator and weigh the empty crucible, W1.

3. Weigh the 2.0g sample into the crucible.

4. Dry the sample in the oven at 100-110°C for 6 hours (dry sample) / 24 hours (wet sample).

5. Cool the sample in desiccator and record the final weight, W2.

Calculation

Percentage of dry sample, % = Weight of dry sample, (W2-W1) g X 100 /Weight of original sample (2g)

Percentage of moisture, % = 100% -percentage of dry sample %

2. Determination of ash content

Method: Ashing

Equipment: Muffle Furnace, 550-600°C, Analytical Balance, Crucible, Desiccator

Procedure:

1. Label and dry crucible in oven at 100-110°C for at least 30minutes.

2. Cool the crucible in desiccator and weigh the empty crucible, W1.

3. Weigh 5.0g sample into the crucible

4. Incinerate the sample in muffle furnace at 550-600°C for overnight

5. Cool the sample in desiccator and record the final weight, W2.

Calculation:

Percentage of ash, % =Weight of ash, (W2-W1) g x 100 / Weight of original sample (5g)

3. Determination of crude fat content

Method: Soxhlet Method

Equipment: Crude Fat Determination System (FOSS, LABTEC ST310), Analytical Balance, Desiccator

Chemicals: Petroleum Ether 40-60°C

Procedure:

Sample Preparation

1. Weigh 1.00 g sample (W1) into filter paper. Fold the filter paper, then put into thimble. "Pre-dry sample if necessary; high moisture content (Dry in oven at 100-110°C for 2 hours)

2. Weigh the dried aluminum cup (W2) and fill in the solvent: 40ml of petroleum ether.

STANDARD OPERATING PROCEDURE FORLABTEC ST310

1. Press the "MAINS" POWER key to switch on the control unit and switch on the cooling unit.

2. Move each knob to BOILING position.

3. Attach thimbles to adapters, fastened the thimbles onto the magnet.

4. Lower the knob to RINSING position (The thimble will now hang just below the condenser level.

5. Insert the filled extraction cups using the cup holder.

6. Lower the handle ensuring that the safety hook engages.

7. The cups are now clamped into the condensers.

8. Start the program on the Control Unit by pressing the Ø key.

9. The buzzer signal starts when the set temperature minus 5°C is reached.

10. Move the extraction mode knobs to BOILING position, then press "TIMER"

11. Buzzer signal starts when the countdown reaches Zero.

12. Move the extraction mode knobs to RINSING position, then press "TIMER"

13. Buzzer signal starts when the countdown reaches Zero.

14. After RINSING, close the condenser valves by turning a quarter turn.

15. Press "TIMER to stat the RECOVERY (Last traces of solvent will be collected in ten

Condenser). *Buzzer signal starts when the recovery countdown reaches Zero.

16. Press "TIMER' to start PRE-DRYING, After PRE-DRYING, take out the cups with the cup holder.

17. Take out thimbles. *CAUTION - BEWARE OF THE HOT PLATE SURFACE.

18. Stop the cooling water supply and turn the "POWER" off.

19. Cups will be further dried in drying oven at 100-110°C for 2 hours.

20. Cool the cups in desiccator and record the final weight (W3).

Calculation:

Percentage of crude fat, % = (W, - W)/ Wix100%

5.0 Determination of crude protein

Method: Kjedahl Method (Digestion, Distillation &Titration)

Equipment: Crude Protein Determination System (digestion &Distillation Unit), Analytical Balance, Manual Titration Set

Chemicals: Kjeltabs Catalyst Cu 3.5, H_2SO_4, 4% Boric Acid, 40%NaOH, Green Bromocresol Indicator, 0.1N

PROCEDURE

1. Weigh 1.0g sample into digestion tube.

2. Add 2 tablet of Kjeltabs Cu 3.5 and 12ml of concentrated H_2SO_4 into the digestion tube.

3. Digest the sample in the digestion unit at 420°C for 40 minutes to 1hour. yneenjaut Glue

(REFER SOPFOR DIGESTION UNIT)

4. Cool the sample, then add 75 ml distilled water and 50ml of 40% NaOH

5. Prepare the receiving solution: 25ml of 4% boric acid with 3-5 drops of green bromocresol indicator in conical flask

6. Place the cooled tube and receiving solution into the distillation unit.

(REFER SOP FOR DISTILLATION UNIT)

7. Conduct the distillation process for 4 minutes. The receiving solution will turn to clear green color after the distillation process.

8. Titrate the receiving solution with 0.1N HCI until it turns to grey color.

Calculation:

Percentage of Protein % = Percentage of Nitrogen x F (Protein factor; 6.25)

Percentage of Nitrogen = (T-B) x Nx14.007x100 / Weight of sample in mg

T = Volume of titration of sample

B = volume of titration for blank (0)

N = Normality of HCI (0.1)

Determination of crude protein:

STANDARD OPERATING PROCEDURE FOR PROTEIN DIGESTION UNIT

(DIGESTOR 2006, FOSS)

Press the main switch to turn on the Digester

↓

The set temperature (420°C) is displayed for 1second, followed by the actual temperature of the Digester

↓

Add the weighed sample, reagent (12ml sulfuric acid) and catalyst (2 tablets of Kjeltabs) into the digestion tubes.

↓

Place the digestion tubes in the tube rack

↓

The heating indicator lamp will flash when the set temperature is reached (420°C)

↓

Place the tube rack with digestion tubes into the digester

↓

Close the digestion tubes with reflux condenser head (attached to water tap)

↓

Turn on the water tap fully for 5 minutes, then slow the water flow

↓

Continue the digestion process until green/blue solution formed in the tubes (30-60 minutes-digestion time depends on the types of sample)

↓

Turn off the power and water tap after complete digestion

↓

Take out the tube rack to cool down the sample

↓

Add 75ml of distilled water to cooled tubes before proceed to Distillation Unit

CAUTION-BEWARE OF THE HOT SURFACE OF THE DIGESTOR DETERMINATION OF CRUDE PROTEINCONTENY STANDARD OPERATING PROCEDURE FOR PROTEIN DISTILLATION UNIT (2100 KJELTEC SYSTEM)

Check the level of alkali in the tank. Fill in the 40% NaOH if necessary

↓

Switch on the power of the Distillation Unit

↓

The set alkali volume and time will be displayed

↓

WARM UP / RINSE THE UNIT

↓

Place the digestion tube (with 20-50ml of dH20) on the left platform and the receiver flask (with

↓

20-50ml of dH20) on the right platform

↓

Close the safety door & turn on the water tap

↓

Open the valve for steam by pressing the MANUAL STEAM KEY

↓

Let the distillation continue according to the set time

↓

Remove the rinsing "digestion tube and receiver flask

↓

SAMPLE DISTILLATION

↓

Place the digestion tube (with first diluted sample) and the receiver flask (with receiver solution) on the platform

↓

Close the safety door

↓

Press the ANALYSE KEY to run a complete distillation cycle

↓

Replace the digestion tube and the receiver flask with the next set and continue the same

↓

"'SAMPLE DISTILLATION" step for all samples

↓

Wipe the unit clean from any spillage

↓

Switch off the power and turn off the water tap

↓

*CAUTON - BEWARE OF THE HOT 'DIGESTION TUBE AFTER EACH COMPLETE DISTILLATION CYCLE WEAR THE HEAT RESISTANT GLOVE TOTAKE OUT THE TUBE

4. Determination of crude Fibre content

Method: Gravimetric Method

Equipment: Crude Fiber Determination System, Oven: 100-110°C, Muffle Furnace; 550600°C, Analytical Balance, Crucible, Desiccator

Chemicals: 1.25% H_2SO_4, 1.25% NaOH

Petroleum Ether &Acetone (for high fat content sample)

*1% HCI (for high carbonate content sample)

Procedure:

1. Dry empty FibreBag and weigh using analytical balance (m1).

2. Weigh 1.0g of sample into the empty FiberBag (m2).Insert FiberBags with glass spacers into carousal. *De-fatting of the sample, esp. if fat content is >10% is required by wash the sample 3 times in 120ml petroleum ether 40-60°C.

3. Washing Phase 1 -Boil the sample in 360ml of 0.13M H_2SO_4 for 30 minutes after start of boiling.

4. Removal of Acids-Rinse 3 times with hot distilled water.

5. Washing Phase 2 -Boil the sample in 360ml of 0.31M NaOH for 30 minutes after start of boling.

6. Removal of Alkali -Rinse 3 times with hot distilled water.

"For de-fatted sample, last step for de-fatting process is required by wash the sample 3 times in 120ml acetone.

7. Dry the FiberBag and empty crucible in the oven for 4 hours at 105°C and cooled in the desiccator.

8. Insert the FiberBag into the pre-dried crucible and weigh together (m3).

9. Incinerate the FiberBag in the furnace for overnight at 550 °C.

10. Cool the crucible containing ash in desiccator and weigh (m4).

Calculation

Percentage of Crude Fiber, %= [(m3- m1)- (m4-m5)] X100/m2

*Same procedure was applied to a blank FiberBag for the blank value, m5

m5 = Blank value = m7-m6 (0.0001)

m6 = Incinerating crucible for blank FiberBag, g

m7= Incinerating crucible and ash for blank FiberBag, g

5. **Determination of carbohydrate content**

Percentage of carbohydrate content will be determined by difference.

CALCULATION

Percentage of Carbohydrate = 100% - (A +B+C+D +E)

A = percentage of moisture content

B = percentage of ash content

C = percentage of crude fat content

D = percentage of crude protein content

E = percentage of crude fiber content

Overall Result

Protein: 8.52%

CHO: 6.41%

Fiber: 24.11%

Fat: 10.46%

Ash: 1.23%

Moisture: 49.275%

pH: 5.38

Water Activity: 0.94

Chapter 7

Nutrient Content

Serving size: 3 nuggets (45 gm.)

Amount Per Serving	Per 100 gm.	Per Serving (45 gm.)	% Daily Value Per Serving
Energy	518.74 kcal	233.43 kcal	
Total Fat	10.46 gm.	4.71 gm.	8.73
Carbohydrate	6.41 gm.	2.88 gm.	0.96
Dietary Fiber	24.11 gm.	10.85 gm.	38.74
Protein	8.52 gm.	3.83 gm.	5.11
Moisture Content	49.28 gm.	22.17 ml	

Chapter 8

Costing and Pricing

Pricing:

We use markup pricing to determine the selling price of our products. The markup percentage is 33.3%, which allows us to calculate the gross profit.

The description of raw material for unit product (240 gm per box)

Item	Price
Banana Peel	RM 5.99
Cabbage	RM 3.99
Green Peas	RM 4.70
Mushroom	RM 3.49
TVP	RM 11.00
Onion	RM 2.99
Garlic	RM 9.92
Ginger	RM 10.97
Bread Crumb	RM 11.95
Tomato Sauce	RM 6.20
Soy Sauce	RM 5.10
Black Pepper	RM 5.00
Chili	RM 24.50
Green Chili	RM 15.99
Bread	RM 4
Cooking Oil	RM 2.15
Salt	RM 0.6

Total = RM 3.94 for 240 gm. (Per Box)

Other Cost Related To The Product :

Type of Cost	**Price**
Packaging	RM 0.5
Printing	RM 0.05
Labor	RM 0.025
Over Head Cost	RM 0.025
Transportation	RM 0.5
Marketing Cost	RM 0.9
Total	RM 2

Calculation of the Price:

Total Cost = Raw Material Cost + Other

=3.94+2

= RM 5.94

Mark up = 33%

Markup amount = Total cost * Mark up

= 5.94 * 0.33

= 1.96

Selling Price = Total Cost + Markup amount

= 5.94 + 1.96

=8.00

(For 100 gm. selling price RM 4)

Chapter 9

Packaging and Labeling

Packaging plays an important role in retaining the shelf life of the product. Besides retaining the shelf life of the product packaging also plays an important role in product branding. So, the packaging of a product is an important aspect in developing a new product

Packaging material used for ECONUGGS :

- Primary Packaging: PLA (Polylactic Acid) film.
- Secondary Packaging: Paperboard.

The primary packaging is PLA (Polylactic Acid) film. It is a biodegradable film that is made up of corn starch or sugarcane. It has a very good barrier property and provides overall consumer appeal. Moreover, this type of packaging ensures a long shelf life.

The secondary packaging is mainly done by paperboard. The paperboard will be laminated with a thin layer of plastic in order to have an overall good appearance. The main advantage of this packaging material is its eco-friendly nature which makes it more sustainable. Also the overall arrangement increases consumer appeal. Though it is susceptible to high pressure but it will not be of any harm to the product related to nuggets as these products are not normally subjected to high pressure.





Chapter 10

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Chapter 11

Appendices










Panasonic





